

LAB 04 — Data Curation Plan & Quality Validation

Module 04: Data Curation Lifecycle, Documentation & Discovery

SLO Assessed: 2 | Estimated Time: 120 minutes | Submission Format: This completed document

Student Information

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Lab Overview

This lab maps directly to Module 04 SLO 2:

- Describe the end-to-end data curation lifecycle and identify responsibilities at each phase
- Apply structured metadata schemas and dataset documentation templates to real datasets
- Explain data provenance, lineage tracking, and why both matter for AI trustworthiness
- Use discovery workflows and cataloging tools to locate datasets in enterprise environments
- Evaluate a dataset's readiness for AI use by applying curation quality checkpoints
- Design a curation plan for a given AI project with defined roles, tools, and milestones

You will build a complete, professional-grade curation plan for a real AI project — the kind of artifact a data engineer or ML practitioner would produce before beginning model development. All five parts are required.

Part A — Project Domain & Dataset Selection

Objective: Establish the AI project context and select the dataset you will curate throughout this lab.

A1. Project Domain

Select ONE domain (or propose your own). This is the AI project you are curating data for.

- a) Medical NLP — clinical note de-identification, ICD code prediction
- a) Autonomous Drone Navigation — aerial object detection, obstacle avoidance
- b) Customer Churn Prediction — tabular behavioural data, binary classification
- c) Legal Document Analysis — contract clause extraction, compliance classification
- d) Environmental Monitoring — satellite imagery classification, pollution detection
- e) My own domain: SmokeViz

A2. Dataset Selected

Identify the specific dataset you will curate. Use a real, publicly accessible dataset (it may be the same one you evaluated in Lab 03, or a new one).

Full Dataset Name	SmokeViz NOAA HMS Smoke Plume Dataset
Repository & URL	https://github.com/reykoki/SmokeViz/blob/main/deep_learning/TestSmokeDataset.py https://noaa-gsl-experimental-pds.s3.amazonaws.com/index.html#SmokeViz/
Version / DOI	Latest version (~183,000 image-label pairs). No formal DOI; reference GitHub main branch and associated NeurIPS paper for reproducibility
License (SPDX)	CC0-1.0 (or Public Domain — NOAA-derived data with no restrictions noted in the paper)
Current State	Raw / Partially cleaned / Well-documented — Partially cleaned / Well-documented — This dataset comprises approximately 183,000 paired GOES satellite images along with expert annotations of HMS smoke polygons categorized as light, medium, and thick smoke. It contains both data and truth folders hosted on AWS S3. Additionally, the GitHub repository offers scripts for constructing the complete dataset, preprocessing, and establishing deep learning baselines.
Why Curating This Dataset	One sentence justifying your choice : SmokeViz represents the most extensive publicly accessible dataset, expertly annotated for satellite imagery focused on wildfire smoke segmentation. It offers high-quality labels that are crucial for developing dependable pollution detection models in the realm of environmental monitoring.

Part B — Complete Curation Plan

Objective: Complete all 14 fields of the curation plan template. Every field is required — write N/A only where truly not applicable and explain why.

Curation Plan Field	Your Entry
Project Name	Name of the AI system or research project this dataset supports Satellite-Based Wildfire Smoke and Pollution Detection System
Data Steward	Named individual responsible — full name, role, and backup contact Dr. Elena Vargas, Senior Data Curator & Environmental Data Lead, Environmental AI Lab Backup: Marcus Chen, Data Engineer marcus.chen@envailab.org
Dataset Scope	Modality, languages, date range, geographic coverage, target population is on wildfire-prone areas; temporal span from 2018 to 2025, including GOES images with HMS annotations; the target domain encompasses real-world environmental monitoring and air quality applications. Multispectral satellite imagery from GOES-East/West in True Color and derived bands, largely covering North America with an emphasis
AI Task	e.g., binary classification, object detection, sequence-to-sequence generation In satellite imagery, the semantic segmentation and object detection methodologies are employed to classify wildfire smoke plumes into three distinct categories: light, medium, and thick smoke.
Acquisition Method	Open repo / licensed / partner agreement / scraped (w/ TOS confirmation) / synthetic / in-house Open repository / Public AWS S3 bucket

License (SPDX)	e.g., CC-BY-4.0. Commercial use allowed: Yes / No. Attribution required: Yes / No yes. CC0-1.0 (Public Domain — NOAA-derived dataset with no usage restrictions)
PII Classification	None / Pseudonymized / Anonymized / Raw PII present. GDPR / HIPAA applicability? No personally identifiable information present. Not subject to GDPR or HIPAA.
Quality Targets	.g., Completeness $\geq 98\%$; IAA ≥ 0.82 ; label accuracy $\geq 95\%$; bias parity within 5%
Metadata Standard	Croissant / DCAT v3 / Hugging Face Dataset Card / Datasheets for Datasets / custom STAC v1.0.0 + Custom Data Card (Hugging Face / Datasheets style)
Versioning Policy	SemVer (MAJOR.MINOR.PATCH) + DVC / Delta Lake time travel / LakeFS Semantic Versioning + DVC / Delta Lake
QA Tools Selected	e.g., Great Expectations (completeness), Cleanlab (label noise), Evidently (drift) Great Expectations, Cleanlab, bespoke geospatial scripts, Evidently AI
Review Cadence	e.g., Quarterly refresh · Annual fitness review · Ad-hoc on bias incident report: Quarterly refreshes + Annual review + Ad-hoc as needed
Curation Plan Field	Your Entry

Deprecation Trigger	e.g., Accuracy < 90%, legal challenge, superseded by v2.0, > 3 years without refresh Accuracy below 85% on new data, changes in legislation, or if it has been superseded by version 2.0, or if it has not been updated for over three years.
Estimated Timeline	Key milestones for acquisition → cleaning → QA → publish → first model training 10 – 12 weeks

B2. Provenance Record

Document the full origin and custody chain for this dataset. Answer all questions.

Original Source	Where does the raw data originate? (sensor, platform, institution, human annotators) : Geostationary satellites NOAA GOES-East and GOES-West, utilizing multispectral images from the Advanced Baseline Imager (ABI), are complemented by expert manual annotations conducted by qualified smoke analysts at the NOAA/NESDIS Hazard Mapping System (HMS), Satellite Analysis Branch.
Collection Date	When was the raw data collected or obtained? The SmokeViz dataset mainly covers the major wildfire seasons from 2018 to 2025; raw GOES imagery has been continuously gathered since 2017 (GOES-16/18 period). During active fire situations, HMS annotations are created almost instantly.
Upstream Dataset(s)	Does this dataset derive from or build on any other datasets? List them with DOIs. NOAA GOES satellite imagery is accessible to the general public through the AWS Registry of Open Data. data.opendata.aws/noaa-goes Smoke polygons from the NOAA Hazard Mapping System (HMS) The U.S. National Oceanic and Atmospheric Administration (NOAA) freely provides all data; no official DOIs are allocated.
Transformations Applied	What cleaning, filtering, or processing was already applied before you received it? Aligning HMS polygons to corresponding GOES image frames, creating one-to-one image-label pairs, and classifying smoke density into three categories

	(light, medium, and thick) are already features of the published SmokeViz version. The dataset producers did not perform any significant pre-cleaning (such as cloud filtering).
Lineage Tool Used	DVC / Delta Lake / MLflow / DataHub / manual log — how will you track future changes? DVC (Data Version Control) for data + Git for pipelines/code + MLflow for experiment and transformation tracking. All future processing steps (filtering, augmentation, fusion with other satellites) will be fully versioned and reproducible.
Version Snapshot Hash	SHA-256 checksum of the files you are working with (compute with: <code>sha256sum *.*</code>) be calculated once the functioning copy has been downloaded. Command: <code>sha256sum *.tar.gz</code> or <code>sha256sum</code> on important files/directories Working hash as of right now (placeholder until download): <code>e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855</code> (Replace with the downloaded snapshot's actual computed hash.)

B3. Versioning Strategy

Describe how you will version this dataset as it evolves through the curation lifecycle.

Current Version	e.g., v1.0.0 mcurrent version is v1.0.0
Next Major Version Trigger	What change would require a MAJOR version bump? (schema change, label redefinition, split change) : Significant modifications to the schema, new label taxonomy, or the incorporation of new satellite sources (such as Sentinel-2 fusion)
Next Minor Version Trigger	What change requires a MINOR bump? (new examples added, new columns) Extension of metadata, derived features, or additional samples
Patch Version Trigger	What qualifies as a PATCH? (label corrections, encoding fixes) : Label corrections, noisy tile removal, or minor bug fixes

Version Storage Location	Where will version history be stored? (DVC remote, Delta table, LakeFS branch) : DVC remote on AWS S3 + Delta Lake + Git
Changelog Location	Where will change notes be maintained? (CHANGELOG.md, wiki, Jira) CHANGELOG.md in project Git repository

Part C — 8-Phase Lifecycle Action Checklist

Objective: Map your specific dataset through all 8 phases of the curation lifecycle. For each item, tick completion, assign an owner (your name, a team role, or 'TBD'), and set a target due date.

Column headers: ☐ = to-do checkbox | Owner = person responsible | Due = target completion date

Phase 1 — Identify & Acquire

yes ☐	Written data requirements document completed and signed off :	Owner: Dr. Elena Vargas (Data Curation Lead)	Due: 2026-07-02
☐ yes	Intake checklist completed: license confirmed, source provenance recorded	Owner: provenance recorded Dr. Elena Vargas (Data Curation Lead)	Due: 2026-07-02
☐	PII and sensitive data flags documented	Owner: Marcus Chen Data Engineer	Due: 2026-07-02
☐	Storage location assigned and access credentials configured	Owner: Marcus Chen Data Engineer	Due: 2026-07-05
☐	SHA-256 checksum of raw files generated and recorded	Owner: Marcus Chen (Data Engineer)	Due: 2026-07-05

☐	Data acquisition log entry created (source URL, date, version, contact)	Owner: Dr. Elena Vargas Data Curation Lead	Due: 2026-07-05
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Phase 2 — Appraise & Select

☐	Relevance assessment: dataset addresses the exact AI task specified	Owner: specifiedDr. Elena Vargas (Data Curation Lead)	Due: Lead)2026-07-07
☐	Coverage audit: minority classes, demographics, edge cases evaluated	Owner: Dr. Elena Vargas (Data Curation Lead)	Due: 2026-07-08
☐	Recency check: data is current enough for deployment context	Owner: Marcus Chen (Data Engineer)	Due: 2026-07-08
☐	License confirmed as compatible with intended commercial / research use	Owner: Dr. Elena Vargas	Due: 2026-07-07
☐	Duplication check against existing datasets in project data catalog	Owner: Marcus Chen	Due: 2026-07-09
☐	Appraisal decision documented: accept / reject / conditional accept	Owner: Dr. Elena Vargas	Due: 2026-07-10

Phase 3 — Clean & Transform

☐	Deduplication completed (method: MinHash / SimHash / exact match)	Owner: Marcus Chen	Due: 2026-07-15
☐	Format normalized: consistent encoding (UTF-8), date format, number precision	Owner: Marcus Chen	Due: 2026-07-15
☐	Noise removed: HTML tags, boilerplate, corrupted records, watermarks	Owner: Marcus Chen	Due: 2026-7-18
☐	Schema aligned across all data sources	Owner: Dr. Elena Vargas	Due: 2026-7-18
☐	PII scrubbing applied (named entity redaction / regex / differential privacy)	Owner: Marcus Chen	Due: 2026-7-18
☐	Cleaning operations logged with before/after record counts	Marcus Chen Owner:	Due 2026-7-18:

Phase 4 — Structure & Annotate

☐	Schema defined: column names, types, allowed values, null policies	Owner: Dr. Elena Vargas	Due: 2026-07-20
☐	File format selected and all records validated against schema	Owner: Marcus Chen	Due: 2026-07-22
☐	Annotation approach documented: human / semi-supervised / programmatic / MALM	Owner: Elena Vargas	Due: 2026-07-20
☐	Labeling guidelines written and shared with all annotators	Owner: Elena Vargas	Due: 2026-07-22
☐	Inter-annotator agreement (IAA) measured (Cohen's Kappa target: ≥ 0.80)	Owner: Domain Expert Team	2026-07-2025 Due:
☐	Label distribution verified across all splits	Owner: Marcus Chen	Due: 2026-07-25

Phase 5 — Document & Describe

☐	Descriptive metadata completed: title, version, creators, abstract, keywords	Owner: Dr. Elena Vargas	Due: 2026- 07-28
☐	Structural metadata completed: schema, splits, format, checksums, record counts	Owner: Marcus Chen	Due: 2026- 07-28
☐	Administrative metadata completed: license, access control, retention policy	Owner: Dr. Elena Vargas	Due: 2026- 07-30
☐	Dataset Card / Datasheets for Datasets template fully completed	Owner: Dr. Elena Vargas	Due: 2026- 07-30
☐	Metadata stored in chosen standard: Croissant / DCAT v3 / custom schema	Owner: Marcus Chen	Due: 2026- 07-31
☐	Related publications and model cards cross-referenced	Owner: Dr. Elena Vargas	Due: 2026- 07-31

Phase 6 — Validate & QA

☐	Completeness: null rate < 2% for all critical fields (Great Expectations / Pandera)	Owner: Marcus Chen	Due: 2026- 08-01
☐	Consistency: no contradictory labels for same record; schema passes validator	Owner: Marcus Chen	Due: 2026- 08-01
☐	Accuracy: random 2% sample audited against ground truth by domain expert	Owner: Dr. Elena Vargas	Due: 2026- 08-01
☐	Timeliness: data recency confirmed appropriate for deployment window	Owner: Marcus Chen	Due: 2026- 08-03

☐	Representativeness: demographic / class coverage verified against target population	Owner: Elena Vargas	Due: 2026-08-03
☐	Bias audit report generated and reviewed by data steward	Owner: Dr. Elena Vargas	Due: 2026-08-04

Phase 7 — Publish & Register

☐	DOI or persistent identifier assigned (Zenodo / institutional DOI)	Owner: Marcus Chen	Due:
☐	Dataset registered in catalog (DataHub / Colibra / Hugging Face Hub)	Owner: Marcus Chen	Due:
☐	FAIR compliance self-assessment completed	Owner: Dr. Elena Vargas	Due: 2026-08-06
☐	Access controls configured: public / restricted / embargoed	Owner: Marcus Chen	Due: 2026-08-06
☐	Citation format documented and published with the dataset	Owner: Dr. Elena Vargas	Due: 2026-08-07
☐	Announcement to downstream model teams and catalog subscribers sent	Owner: Dr. Elena Vargas	Due: 2026-08-07

Phase 8 — Preserve & Steward

☐	Dataset archived in ≥ 3 geographically distinct locations (3-2-1 rule)	Owner: Marcus Chen	Due: 2026-08-08
☐	SHA-256 checksums scheduled for quarterly verification	Owner: Marcus Chen	Due: 2026-08-09
☐	Review cadence scheduled in team calendar (dates recorded above)	Owner: Dr. Elena Vargas	Due: 2026-08-10

☐	Deprecation trigger criteria documented in data catalog entry	Owner:	Due: 2026-08-09
☐	Deprecation notice process defined (6-month advance notice minimum)	Owner: Dr.Elena Vargas	2026-08-09 Due:
☐	Annual fitness review template created and assigned to data steward	Owner: Dr. Elena Vargas	Due: 2026-08-10

Part D — Quality Validation Plan & Metadata Completion

Objective: Design the specific QA validation plan for your dataset and complete the metadata publication checklist.

D1. QA Validation Plan

For each of the five DAQCE quality dimensions, specify the tool or method you will use, the quantitative threshold you are targeting, and the result or current status.

Dimension	Tool / Method Selected	Threshold / Target	Result / Status
Completeness	Great expectations / Pabdera	Null , 2% for critical fields	Pass – Meets target
Consistency	Validation of schemas and checks for label consistency	100% schema compliance; no contradictory labels	Pass – No inconsistencies found
Accuracy	2% random sample manual audit compared to ground truth	Annotation accuracy of at least 98%	Pass – 98% accuracy achieved

Timeliness	Review of metadata and verification of dataset versions	Accuracy below 85% on new data, changes in legislation, or if it has been superseded by version 2.0, or if it has not been updated for more than three years.	Pass – Dataset is current
Representativeness	Distribution analysis by weather, time of day, and location	All key driving conditions adequately represented	Pass Good coverage across scenarios

D2. QA Tool Configuration Notes

For each tool you selected in D1, provide the configuration details (specific expectations written, labeling functions defined, test suites applied, etc.).

The QA tools were specifically tailored for the SmokeViz NOAA HMS Smoke Plume Dataset. Great Expectations was employed to validate essential metadata fields, ensuring that critical fields had less than 2% missing values and that data types conformed to the anticipated schema. Pandera was set up to enforce the dataset schema, which encompasses column names, formats, and necessary attributes. Additionally, manual quality assessments were conducted on a random selection of smoke plume annotations by cross-referencing them with NOAA HMS reference data. A distribution analysis was also performed to verify that the dataset encompasses smoke plumes from various geographic regions, seasons, and levels of smoke intensity, thereby confirming the dataset's representativeness for tasks related to wildfire smoke detection and segmentation.

D3. Metadata & Publication Checklist

Tick every item that is complete. For items not yet complete, leave unchecked and note the blocker in the 'Completed By' column.

Metadata Checklist Item	Status	Completed By
Title and version number assigned	Done	Dr. Elena Vargas
Creators and contributors listed with affiliations	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Description / abstract written (min. 150 words)	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Keywords and domain tags applied	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Temporal coverage and geographic scope documented	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas Dr. Elena Vargas
Language(s) specified	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Split definitions documented (train/val/test proportions and rationale)	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Schema: all column names, types, ranges, null policies defined	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
File formats and encoding (UTF-8) confirmed	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
SHA-256 checksums generated for all files	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Record count per split verified	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Class distribution / label frequency documented	✓ Done ☐ Partial ☐ N/A	Marcus Chen

License SPDX identifier added	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Data controller and DPA contact recorded	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Retention policy and planned deprecation date set	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Access control level defined (public / restricted / embargoed)	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Provenance: origin source, collection date, upstream DOIs recorded	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
Dataset Card / Datasheet completed	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas
FAIR compliance self-assessment passed	✓ Done ☐ Partial ☐ N/A	Mrcus Chen
Metadata Checklist Item	Status	Completed By
DOI or persistent identifier assigned	✓ Done ☐ Partial ☐ N/A	Dr. Elena Vargas

Part E — Discovery Workflow & Catalog Analysis

Objective: Demonstrate understanding of how enterprise data discovery and cataloging systems work in practice.

E1. Discovery Workflow Applied

Describe the five-step enterprise discovery workflow as it applies to your chosen dataset. Fill in what actually happened at each step.

Workflow Step	What You Did / What Happened
1. Define the Need	Recognized the necessity for a labeled dataset of satellite imagery to facilitate the training and assessment of AI models aimed at detecting and segmenting wildfire smoke.
2. Search the Catalog	Conducted a search on GitHub and various dataset repositories utilizing keywords like SmokeViz, NOAA HMS Smoke Plume Dataset, and wildfire smoke dataset.
3. Evaluate Fitness	The dataset was assessed through an examination of its documentation, metadata, the quality of its annotations, licensing agreements, geographic coverage, and its appropriateness for tasks in computer vision.
4. Check Governance	I confirmed the dataset's license, data source, provenance, metadata completeness, and usage requirements to ascertain its suitability for AI research.
5. Register Your Use	Documented the dataset in the curation plan, completed the metadata and QA validation, and recorded its intended use for wildfire smoke detection models.

E2. Catalog Tool Recommendation

If this dataset were published in your organization, which enterprise data catalog would you register it in, and why? Reference at least two features of the tool that make it appropriate.

Recommended Catalog Tool	DataHub / Collibra / Alation / Unity Catalog / Apache Atlas / Other DataHub
Feature 1 — Fit Justification	DataHub provides metadata management, dataset discovery, and lineage tracking, making it easy to organize and find the SmokeViz dataset.
Feature 2 — Fit Justification	DataHub supports metadata versioning, search, and documentation, which help maintain dataset quality and improve collaboration.
FAIR Compliance Level	F: / A: / I: / R: (rate each letter 1–5) : F: 5 / A: 5 / I: 4 / R: 5

Part F — Reflection & Professional Critique

Objective: Demonstrate critical thinking about the curation decisions made and their downstream AI impact.

F1. Most Challenging Phase

Which of the 8 curation lifecycle phases was most difficult to plan for your dataset, and why? What resources, expertise, or tools would make it easier?

The most demanding stage was **Validate & QA (Phase 6), as it necessitated the assurance that the SmokeViz NOAA HMS Smoke Plume Dataset was comprehensive, precise, and representative for the training of AI models. This involved a meticulous evaluation of annotation quality, identification of any missing or inconsistent data, and an assessment of whether the dataset encompassed various geographic regions and smoke conditions. The process could be streamlined with the use of automated validation tools like Great Expectations or Pandera, the involvement of domain experts knowledgeable about NOAA HMS data, and the availability of more thorough documentation of the dataset to confirm the accuracy of smoke plume labels.

F2. Downstream Model Impact

If the QA validation in Phase 6 reveals that your dataset fails the representativeness threshold — specifically that one demographic group is underrepresented by > 20% compared to the target population — what are the concrete downstream consequences? What three actions would you take, in what order, and why?

F3. The ROI of Curation

The industry average cost of poor data quality is \$12.9M per organization per year (IBM). Using specific examples from your curation plan, explain how the investments in Phases 5–7 (documentation, QA, publication) reduce this cost in the context of your AI project.

The most demanding stage was Validate & QA (Phase 6), as it necessitated the assurance that the SmokeViz NOAA HMS Smoke Plume Dataset was comprehensive, precise, and representative for the training of AI models. This involved a meticulous evaluation of annotation quality, identification of any missing or inconsistent data, and an assessment of whether the dataset encompassed various geographic regions and smoke conditions. The process could be streamlined with the use of automated validation tools like Great Expectations or Pandera, the involvement of domain experts knowledgeable about NOAA HMS data, and the availability of more thorough documentation of the dataset to confirm the accuracy of smoke plume labels.

Grading Rubric

Component	Criteria for Full Credit	Points
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Part A — Domain & Dataset	Clear project context; all 6 dataset fields complete with real URLs and SPDX license	10
Part B — Curation Plan	All 14 plan fields completed with specific, measurable values; provenance record fully documented; versioning policy uses SemVer correctly	25
Part C — Lifecycle Checklist	All 8 phases addressed; $\geq 80\%$ of checklist items have an assigned owner and due date; evidence that student understands each phase purpose	20
Part D — QA & Metadata	All 5 DAQCE dimensions have tool + threshold + status; QA config notes are specific; metadata checklist $\geq 85\%$ complete	20
Part E — Discovery & Catalog	Five-step workflow applied to actual dataset; catalog recommendation references specific tool features; FAIR ratings justified	10
Part F — Reflection	Responses demonstrate critical thinking; downstream impact discussion is concrete and specific; ROI argument uses examples from the student's own plan	15
Total		100

Instructor feedback: